

STATIONARITY AND THE STATIONARY DISTRIBUTION OF A CERTAIN MARKOV CHAIN IN INVENTORY CONTROL

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Key words: Stationarity, Stationary Distribution, Inventory, Markov Chain.

1. Introduction

Define a matrix $\{P_{ij}\}$ by

$$(1) \quad P_{ij} = \begin{cases} r_{s-i} & \text{for } i = 0, 1, \dots, s \\ 0 & \text{for } i = s+1, s+2, \dots \end{cases}$$

while for $j=1, 2, \dots$

$$P_{ij} = \begin{cases} p_{s+j-i} & \text{for } i = 0, 1, \dots, s+j \\ 0 & \text{for } i = s+j+1, s+j+2, \dots \end{cases}$$

where $p_i \geq 0$, $i = 0, 1, 2, \dots$; $\sum_{i=0}^{\infty} p_i = 1$; $r_k = \sum_{i=0}^k p_i$ and s is some finite number.

This matrix can be regarded as the matrix of one-step transition probabilities of a Markov chain, and is well known in the literature as the transition matrix in the infinite discrete dam problem (see Prabhu (1965), p. 163), and in queueing theory with bulk service in connection with the length of the queue at epochs just before service (Bailey (1954) and Gani and Prabhu (1957)).

In this study, the matrix occurred in an inventory problem, governing the transitions of the opening inventories of cycles (to be discussed in paragraph 4).

The problem is twofold: to find conditions for the system to be positive recurrent (ergodic), null recurrent or transient; and to find the stationary distribution of the Markov chain, if it exists.

A short method is proposed by which null recurrency may be deduced in some cases. Furthermore, the stationary probabilities themselves were needed in the problem from which (1) was derived.

Therefore, an "in principle"-type of solution was not sufficient. Four methods by which the stationary distribution may be derived, will be discussed.

2. Conditions for Stationarity

Assume $0 < p_0 < 1$ and $r_s < 1$. Then the system is aperiodic and irreducible. Hence, all states are of the same type (Feller (1968), p. 391). That is, every state is null recurrent or every state is transient and

$$\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0 \text{ for all } i \text{ and } j,$$

or every state is positive recurrent and $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = u_j > 0$ for all i and j , where $P_{ij}^{(n)}$ represents the n -step transition probability.

$$\text{Define } P(z) = \sum_{j=0}^{\infty} z^j p_j, \text{ and let } \nu = P'(1) = \sum_{j=0}^{\infty} j p_j$$

(i) Assume $s < \nu$. Let $0 < i < s$ and consider the system of inequalities

$$(2) \quad \sum_{j=0}^{\infty} P_{ij} y_j \leq y_i$$

with $y_j = \alpha^j$, $j=0,1,2,\dots$, $0 < \alpha < 1$. This is equivalent to

$$r_{s-i} + \sum_{j=1}^{\infty} p_{s+j-i} \alpha^j \leq \alpha^i$$

which, after some manipulation, reduces to

$$(3) \quad \sum_{k=0}^{s-i} p_k (\alpha^{s-i-k} - \alpha^k) + P(\alpha) \leq \alpha^s.$$

If the existence of an α , $0 < \alpha < 1$, can be shown, satisfying

$$(4) \quad P(\alpha) \leq \alpha^s,$$

then, since $(\alpha^{s-i-k} - \alpha^k) \leq 0$ for $k \leq s-i$, (3) will be satisfied by the same α , and hence (2) will be satisfied by the same α .

Since by assumption, $P(0) = p_0 > 0$, and $s < \nu$, it can easily be shown that an α_0 , $0 < \alpha_0 < 1$, and satisfying (4), does exist.

For $i \geq s$, (2) immediately leads to (4). Thus, for all $i \neq 0$, the system (2) is satisfied by $y_j = \alpha_0^j$, $j=0,1,2,\dots$.

According to Foster (1953), theorem 6, the system is transient.

- (ii) Assume $s > \nu$, that is, $s - \nu \geq \epsilon > 0$. Then, for $i < s$ and $y_j = j, j=0,1,2,\dots$ (whence $y_j \rightarrow \infty$),

$$\sum_{j=0}^{\infty} y_j P_{ij} = \sum_{j=1}^{\infty} j p_{s+j-i} \leq \nu - (s-i) \sum_{j=s-i}^{\infty} p_j < \infty.$$

Furthermore, for $i \geq s$:

$$\sum_{j=0}^{\infty} y_j P_{ij} = \sum_{j=0}^{\infty} (j+i-s)p_j \leq \nu + i - s \leq y_i - \epsilon.$$

Dividing by ϵ and using Theorem 1 of Pakes (1969), it follows that the system is positive recurrent.

- (iii) In considering null recurrency, it will be shown that the system is recurrent (i.e. positive or null recurrent) for $s \geq \nu$. For, with $y_j = j$

$$\sum_{j=0}^{\infty} y_j P_{ij} = \sum_{j=0}^{\infty} (j+i-s)p_j = \nu + i - s \leq y_i, \text{ where } i \geq s.$$

From Pakes (1969), theorem 3, it follows that the system is recurrent for $\nu \leq s$, and, in fact, from (ii) positive recurrent for $\nu < s$.

It will now be shown that the system cannot be positive recurrent for $s = \nu$, and thus it must be null recurrent in this case.

To this end, assume the system to be positive recurrent for $\nu = s$. Then the stationary probabilities $\{u_k\}$ exist (Feller (1968)), with $u_k > 0$ and $\sum u_k = 1$, satisfying

$$(5) \quad u_j = \sum_i u_i P_{ij}.$$

From the probability generating function

$$U(z) = \sum_{j=0}^{\infty} z^j u_j$$

it follows, by (5) and some manipulation that

$$U(z) = z^{-s} U(z) P(z) + \sum_{i=0}^s v_i (1 - z^{i-s})$$

where $v_k = \sum_{j=0}^k u_j p_{k-j}$ - the convolution of $\{u_k\}$ and $\{p_k\}$.

Hence

$$(6) \quad U(z) = \frac{\sum_{i=0}^s v_i (z^s - z^i)}{z^s - P(z)}$$

Since $\sum u_k = 1$, it must be true that $U(1) = \sum u_k = 1$.

But

$$\begin{aligned}\lim_{z \rightarrow 1^-} U(z) &= \lim_{z \rightarrow 1^-} \frac{\sum v_i (z^s - z^i)}{z^s - P(z)} \\ &= \lim_{z \rightarrow 1^-} \frac{\sum v_i (sz^{s-1} - iz^{i-1})}{sz^{s-1} - P'(z)} \\ &\quad \text{by L'Hospital's rule.}\end{aligned}$$

$$= \infty \text{ because } i \leq s \text{ and } \nu = s.$$

This contradicts the assumption of positive recurrency, and thus the system is null recurrent when $\nu = s$.

We have the following

Theorem

The Markov chain with transition probability matrix (1), is

<i>transient if and only if</i>	$s < \nu$
<i>positive recurrent if and only if</i>	$s > \nu$
<i>null recurrent if and only if</i>	$s = \nu$

where $\nu = \sum j p_j$.

3. The Stationary Distribution

In the positive recurrent case the stationary distribution is known to exist (Feller (1968)). A well-known "in principle"-type of solution will be given, as well as three methods by which the stationary probabilities can be computed numerically.

(i) By analysing $U(z)$ still further, analogous to Bailey (1954), one finds that

$$U(z) = \frac{(s-\nu)(z-1) \prod_{i=1}^{s-1} \left(\frac{z-z_i}{1-z_i} \right)}{z^s - P(z)}$$

where $z_1, \dots, z_{s-1}$ are the $s-1$ roots of $z^s - P(z)$ inside the unit circle.

Sufficient conditions for the existence of the $s-1$ roots are:

$$(a) \quad \nu < s$$

$$(b) \quad \overline{\lim}_{n \rightarrow \infty} n \sqrt{p_n} < 1.$$

$U(z)$ may then be expanded in a power series in z and the probabilities $\{u_i\}$ can be obtained in the usual way. Unfortunately, the power series expansion is not a trivial problem.

- (ii) Since $U(z)$ and $P(z)$ are analytic for $z \leq 1$, the s zeros of $\sum v_i(z^s - z^i)$ and $z^s - P(z)$ must coincide inside and on the unit circle. (Ahlfors (1953)). Let these zeros be $z_0 = 1, z_1, \dots, z_{s-1}$. Since $U(z_0) = 1$, the numerator and denominator of (6) may be differentiated with respect to z_0 (according to L'Hospital) and the result set equal to 1. This gives

$$(7) \quad \sum_{i=0}^{s-1} v_i(s-i) = s-\nu$$

It must also be true that

$$(8) \quad \sum_{i=0}^{s-1} v_i(z_j^s - z_j^i) = 0 \text{ for } j=1, \dots, s-1.$$

Hence we arrive at the following system of equations:

$$\begin{aligned} (9) \quad & v_0^s + v_1(s-1) + \dots + v_{s-1} = s-\nu \\ & v_0(z_1^s - 1) + v_1(z_1^s - z_1) + \dots + v_{s-1}(z_1^s - z_1^{s-1}) = 0 \\ & \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \\ & v_0(z_{s-1}^s - 1) + v_1(z_{s-1}^s - z_{s-1}) + \dots + v_{s-1}(z_{s-1}^s - z_{s-1}^{s-1}) = 0 \end{aligned}$$

where z_k is a root of $z^s - P(z)$ within $z=1, k=1, \dots, s-1$. By substituting

$$z_k = \gamma_k (\cos \theta_k + i \sin \theta_k)$$

in (9), it can be split into a real and an imaginary system of equations, of the form

$$(10) \quad A(\underline{\gamma}, \underline{\theta}) + i B(\underline{\gamma}, \underline{\theta}) = 0.$$

For (10) to hold, it is necessary that

$$A(\underline{\gamma}, \underline{\theta}) = \underline{0}.$$

This last equation can be solved for v_i , $i = 0, \dots, s-1$. Then u_i , $i = 0, \dots, s-1$, can be found from the definition of v_i , namely $v_i = \sum_{j=0}^{i-1} u_j p_{i-j}$, $i = 0, \dots, s-1$. The rest of the u_i 's can be computed through the relationship

$$u_j = \sum u_i p_{ij}.$$

This method can be applied if one can find the roots of

$$(11) \quad z^s - P(z)$$

with ease. Again with the transformation $z_k = \gamma_k (\cos \theta_k + i \sin \theta_k)$, (11) can be written in the form $f(\gamma_k, \theta_k) + i g(\gamma_k, \theta_k) = 0$, from which γ_k and θ_k can be computed as solutions of

$$f(\gamma_k, \theta_k) = 0 \text{ and } g(\gamma_k, \theta_k) = 0.$$

For the case $P(z) = e^{-\nu(1-z)}$, which was actually considered, the two-variable Newton-Raphson method was used to find the roots. For small values of s this was not too difficult, but for large s the different zeros got very close to one another, making the Newton-Raphson method rather sensitive with respect to the initial guess, and the work therefore tedious.

This reduces the applicability of the method.

(iii) This stationary probabilities $\{u_i\}$ can always be calculated

$$\text{as } u_j = \lim_{n \rightarrow \infty} P_{ij}^{(n)} \quad (\text{Feller (1968), p. 393}).$$

An iterative method, which is less time-consuming, easily programmable, and which can be used on a small computer, is based on the following:

Define $\underline{U}^{(0)} = (1 \ 0 \ 0 \ \dots)$ and $\underline{U}^{(n)} = \underline{U}^{(n-1)} P$, where P is the transition probability matrix. Then, since $u_j = \lim_{n \rightarrow \infty} P_{0j}^{(n)}$ we get by iterating on the recursive relationship above, $u_j^{(n)} = P_{0j}^{(n)}$, and hence $\underline{U}^{(n)} \rightarrow \underline{U} = (u_0 \ u_1 \ \dots)$.

Note that the transition matrix need not have the special form of (1).

- (iv) Still another method can be used, utilising the special form of (1). According to Bharucha-Reid (1960), theorem 1.6, there exists a unique solution $\{u_k\}$, with $u_k > 0$ and $\sum u_k = 1$, to the system of equations (5), which, when written out in this particular case, becomes

$$(12) \quad u_0 = u_0 r_s + u_1 r_{s-1} + \dots + u_s r_0$$

$$u_i = u_0 p_{s+i} + u_1 p_{s+i-1} + \dots + u_{s+i} p_0, \quad i = 1, 2, \dots$$

By adding the second row and the first, then the third, second and first, etc., and using the transformation to cumulative distribution functions

$$(13) \quad y_i = \sum_{j=0}^i u_j, \quad i = 0, 1, 2, \dots,$$

(12) becomes

$$(14) \quad y_i = p_{s+i} y_0 + p_{s+i-1} y_1 + \dots + p_0 y_{s+i}, \quad i = 0, 1, 2, \dots$$

If a solution to (14) can be found, (12) can be solved by (13).

Define

$$(15) \quad y_i^{(n+1)} = p_{s+i} y_0^{(n)} + \dots + p_0 y_{s+i}^{(n)}, \quad i = 0, 1, 2, \dots,$$

with initial solution $y_i^{(0)} = 1$ for $i = 0, 1, 2, \dots$.

It can easily be shown that this iteration procedure converges monotonically to the solution of (14).

The iteration method can be accelerated by using in stead of (15), the system

$$y_i^{(n+1)} = p_{s+i} y_0^{(n+1)} + \dots + p_{s+1} y_{i-1}^{(n+1)} + p_s y_i^{(n)} + \dots + p_0 y_{s+i}^{(n)}, \quad i = 1, 2, \dots$$

where a negative subscript indicates that that particular term must be deleted. It can be shown that this last system converges at least as fast as (15).

The recurrence system can be programmed easily and convergence is fast. This last

method is the most effective for obtaining the stationary probabilities related to (1), in the case $P(z) = e^{-\lambda(1-z)}$.

4. Examples

- (i) Consider the following inventory system: Demand for a certain item follows an exponential interarrivaltime distribution with expected value $1/\lambda$, units being demanded one at a time. Then demand per unit of time follows a Poisson distribution with expected value λ . Demands, occurring when the system is out of stock, are back-ordered to be delivered when an order arrives. An order is delivered into stock, a constant time τ timeunits after being placed. The following decision rule holds:

If nett stock (stock on hand minus backorders) immediately after the delivery of an order, is more than R units, an order for Q units is placed the moment the stock level reaches R units. If, on the other hand, it is less than or equal to R units, Q units are ordered immediately.

Let a cycle be the time interval from the placing of an order until the placing of the following order, and divide it into two phases: phase I the leadtime with length τ , and phase II the period from τ until the next order is placed. (Note that from the decision rule above, phase II may have length zero).

Let X_j denote the amount of stock on hand at the start of the j -th cycle. Then X_j can take on a value from the list $\{ \dots, -1, 0, 1, \dots, R \}$. It can be shown that $\{ X_j \}$ constitutes a Markov chain. Let $P_{jk} \equiv \text{Prob} [X_{n+1} = R-k | X_n = R-j]$. Then P_{j0} is the probability that, given that the previous cycle ended with $(R-j)$ units in nett inventory, the present cycle ends with R units nett inventory; i.e. that there are R or more units left at the end of phase I of the present cycle; i.e. that $Q-j$ or less units have been demanded during a time interval of length τ , in the present cycle.

$$\text{Let } p_k = (\lambda\tau)^k \exp(-\lambda\tau)/k! , \quad k = 0, 1, 2, \dots$$

$$\text{and } r_k = \sum_{i=0}^k p_i , \quad k = 0, 1, 2, \dots$$

$$\begin{aligned} \text{Then } P_{j0} &= r_{Q-j} & \text{for } j &= 0, 1, \dots, Q \\ &= 0 & \text{for } j &> Q. \end{aligned}$$

Similarly, it can be shown that, for $k > 0$

$$\begin{aligned}
 P_{jk} &= P_{Q+k-j} & \text{for} & & j &= 0, 1, \dots, Q+k \\
 &= 0 & \text{for} & & j &> Q+k.
 \end{aligned}$$

Hence the matrix of one-step transition probabilities of the Markov chain $\{X_j\}$ is of the form of (1), and the results of the previous paragraphs apply. From § 2 it follows that the inventory system is

transient		$\lambda\tau > Q$
null recurrent	according to	$\lambda\tau = Q$
positive recurrent		$\lambda\tau < Q$.

Note that stationarity is not influenced by the value of R . We can give the following intuitive meaning to the above conditions: The expected amount demanded during a time interval of length τ , is $\lambda\tau$. The amount by which the stock increases during the same time interval, is Q . No increase in stock can take place during phase II. The system is in steady state or equilibrium, that is, positive recurrent, if the inflow Q is more than the expected outflow $\lambda\tau$. This ensures that the system does not "run empty" — it bounds the system from below. On the other hand, if there are too many items in the system at the end of phase I, phase II ensures that there will be "over-flow", emptying the system until only R units are left. This bounds the system from above.

- (ii) The following example was suggested by the referee: In the previous example, let the successive leadtimes $\tau_1, \tau_2, \dots$ be independent, identically distributed random variables, with distribution function F . Then, in the previous notation:

$$\begin{aligned}
 p_k &= \int_0^\infty (\lambda t)^k e^{-\lambda t} dF(t)/k! \\
 \nu &= \sum_j j p_j = \lambda \int_0^\infty t dF(t) = \lambda E[\tau_1] \\
 P(z) &= \int_0^\infty \exp(-\lambda t(1-z)) dF(t)
 \end{aligned}$$

If the moment generating function exists in the vicinity of the origin, the previous theory applies. In the positive recurrent case, i.e. $\lambda E[\tau_1] < Q$, the stationary distribution can sometimes be computed explicitly. For instance, if $F(t) = 1 - e^{-\mu t}$, by proceeding analogous to Bailey (1954) and using the above, one finds

$$u_0 = (1 + \lambda E[\tau_1])(z_Q - 1) z_Q^{-1}$$

$$u_i = z_Q^{-i} (u_0 - (z_Q - 1) \lambda E[\tau_1]) \quad i = 1, 2, \dots$$

where z_Q is the zero of $z^Q/P(z) - 1$ outside the unit circle. For other cases in the family of gamma distributions, see Bailey (1954).

5. Acknowledgement

The author wishes to thank Prof. J.H. Venter for his assistance and encouragement during all stages of the study on which this paper was based. He would also like to thank the referee for valuable suggestions for improving the paper.

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Manuscript received 3rd May 1972; revised 13th September 1972.